

Roll No.

TCS-351

B. TECH. (CSE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

FUNDAMENTAL OF CLOUD COMPUTING AND BIG DATA

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Using AWS, Solaris built a real-time monitoring application for oilfield equipment. Solaris automated and integrated data ingestion to create a safer, more efficient and cost effective solution for customers. Identify and discuss the architecture of Client/Server Computing model from the perspective of Solaris. (CO1)

OR

- (b) Boost Commerce has maintained 99 percent or higher uptime and scaled to serve 11,000 merchants in its first four years of operations on AWS. Boost Commerce offers software for Shopify merchants to help them improve product discoverability and analytics on their e-commerce pages. Analyze the applicability of the Cloud Deployment Models from the perspective of Boost Commerce. (CO3)

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2. (a) American Airlines wanted to provide convenient digital services for customers and understood there was an opportunity to remove the constraints of the existing legacy architecture, platform, organization, development and operations approaches. Customer-facing applications were based on monolithic code, duplicated and managed in silos. Every change required the same work in up to three places, each managed by different teams. Analyze Web Development on cloud computing from perspective of American Airlines. (CO2)

OR

- (b) FanDuel Group, the leading sports gaming company in the United States, has accelerated its business expansion and improved its user experience by standardizing on Amazon Web Services (AWS) across both its cloud and on-premises operations using AWS Outposts. Critically analyze user Interface (UI) on cloud computing from perspective of FanDuel Group. (CO2)

3. (a) As per NIST, evaluate the characteristics, stakeholders and major service providers that are enabling cloud computing. (CO2)

OR

- (b) Financial technology company Chime removed scale-related downtime to migrating to AWS, providing millions of members with a better experience and freeing time for Chime engineers to innovate. Analyze Cloud Service Models from perspective of Chime. (CO3)

(3)

4. (a) Virtualization is a major enabler for cloud computing. Discuss the role of virtualization in achieving the vision of cloud computing. (CO3)

OR

- (b) "The power of 'open source and standards' is the power of the people and the people always rule." In this context, present a case outlining the need and benefits of using open source and standards ? Distinguish between open source and open standards with appropriate examples. (CO2)

5. (a) Critically assess the abstract interaction dynamics between IaaS, PaaS and SaaS delivery models with appropriate examples. (CO2)

OR

- (b) Outline the evolution of cloud computing. According to you, what are the key milestones for achieving this vision ? (CO3)